

AUTOMATION PROJECT OVERVIEW

This project is to develop a computerized system to help restaurant personnel coordinate their activities and improve their services, and for the management to track business growth and create future plans.

The goal of this project is to introduce automation in privately owned restaurants, that is, small- to-medium sized establishments. Typical problems restaurant personnel are facing include:

- Coordination of their work activities
- Anticipating and handling periods of low/high Customer traffic
- Recognizing trends early enough to take advantage of bestsellers or abandon the flops.
- Lowering operating costs, and increasing efficiency/productivity and profits

This project is to computerize restaurant operations so that all information pertaining to customers' orders and staff activity will be conveniently shared and stored over the restaurant's intranet. Hosts will be able to view table status with a click of a button. The wait staff will be able to enter the customers' orders quickly and efficiently and then have them delivered electronically to the kitchen. The kitchen staff will be able to view the incoming orders and notify the proper wait staff when the food is ready. Busboys will be able to view real-time floor status allowing them to know which tables are clean, dirty, or occupied.

Most importantly, all the restaurant information is organized and saved for the management viewing and archival. The analysis will consist of by-the-day and by-the-hour breakdowns of:

- Revenue and revenue percentage per menu item
- Menu item popularity
- Personnel efficiency
- Average turnaround time (how long customers spend in the restaurant)
- Average preparation time (time when the order is placed to when it is ready)

All the data is automatically collected and processed, allowing management to focus on analyzing the data rather than calculating it.

Throughout the restaurant there will be computer terminals for the staff to log in. The system requires each user to login, complete their task and logout. By using a touch screen the restaurant staff can quickly and efficiently log in and complete the desired task.

The supported employee roles are Waiter, Cook, Busboy and Manager. Every user account in the system should have its own privileges. All the role-personalized home screens for each employee will be refreshed automatically when needed.

The roles and privileges are as follows:

1. WAITER / WAIT-STAFF

When a waiter logs in

- They are greeted with a floor status screen on which their assigned tables are colored.

- Their tables are colored according to status; green is open, yellow is occupied; red is dirty.
- At this point a waiter can select a table to view its tab.
- Once a table is selected, the waiter can choose from several options.
 - If they select to add an item to the table's tab, they are presented with various categories of food items offered. Here they can select the appropriate category and then find the desired item.
 - For example, if a customer ordered a Caesar salad, the waiter would login, select the table, and choose "Add Item." They would then select the "Soups/Salads" from the category list, and then select the desired salad from the items presented. They are then returned to that table's screen where they can choose to perform another task or logout.

2. KITCHEN STAFF:

- View orders placed by wait-staff using the computer terminals on the restaurant floor are displayed to the kitchen staff through a queue, i.e., on a first-in, first- out basis.
- Notify Wait-staff when the food is ready.

3. BUSBOY:

- Busboys should be able to view real-time floor status allowing them to know which tables are clean, dirty, or occupied.
- They should also be able to change the status of the table as needed.

4. MANAGER:

The Manager should have administrative power over all the employee profiles:

- The ability to create, delete and modify profiles, track employee activities, and authorize restricted waiter activities.
 - If the employee is a waiter, his/her profile also contains information about the tables for which he/she is responsible. From this profile, the individual tabs for those tables can be accessed.
- The manager should also have the ability to manage other aspects of restaurant operations, such as inventory tracking and sale analysis.

REMEMBER,

- The standard table layout and Menu (which is included in the Automation Project Menu, Hours, and FAQ document) should be used for this Project.
- Every order should include a unique id number, the server's id, and the table id. The order should show the items that have been requested by each individual seat. There should be a capability to add items to the order, e.g., a dessert, a side, or a beverage.
- In addition to staff coordination, the system tracks everything electronically then organizes it.
- Employee hours are kept, allowing for rapid processing of the payroll.
- Revenue is tracked by day, week, or month. All this information is collected, saved, and then entered into a table format for easy reading by the management. The automatically generated statistics allow the management to

see what portion of the revenue comes from what item, i.e., what are the most popular items.

- All this is done automatically and stays up to date with the restaurant performance.

The developer(s) should count the number of clicks/keystrokes that are necessary to accomplish individual tasks. Make every effort to reduce the number of clicks/keystrokes in the system interaction, while not compromising functionality and security.

As this project develops, there should be many questions that arise. Do not make assumptions about the information or the requirements. The restaurant has a project manager/point of contact (the Instructor) to answer all your questions.